Unit 1

Artificial Intelligence -

Artificial Intelligence (AI) is a technology that enables machines and computers to perform tasks that typically require human intelligence. It allows systems to learn from data, recognise patterns, and make decisions to solve complex problems.

- Simulates human intelligence through learning and reasoning.
- Processes large amounts of data to identify patterns and insights.
- Continuously improves performance using experience and feedback.

Core Concepts

1. Machine Learning (ML)

Machine Learning is a subset of AI that enables systems to learn from data and improve performance without explicit programming.

- Identifies patterns and relationships from data.
- Improves predictions and decisions through experience.
- Powers applications like recommendation and prediction systems.

2. Generative AI

Generative AI focuses on creating new content such as text, images, audio, and videos using learned patterns from large datasets.

- Generates original content instead of only analyzing data.
- Learns patterns from massive datasets for content creation.
- Used in chatbots, image generation, and content writing tools.

3. Natural Language Processing (NLP)

Natural Language Processing (NLP) enables computers to understand, interpret, and generate human language naturally.

- Supports tasks like translation, sentiment analysis, and chatbots.
- Combines linguistics with computer science techniques.
- Enables voice assistants such as Siri and Alexa.

4. Expert Systems

Expert Systems simulate human decision-making using predefined rules and domain-specific knowledge.

- Uses if-then rules to solve specialized problems.

- Mimics reasoning of human experts in specific domains.
- Commonly applied in healthcare, finance, and diagnostics.

Working

1. **Data Collection**: AI systems rely on large sets of data which could include images, text or sensor readings. For example, teaching an AI to recognize cats, we collect a dataset of labeled cat images.
2. **Processing and Learning**: It uses algorithms to analyze data and identify patterns. For example, it learns to recognize key features like a cat's shape, ears or whiskers helping it understand the data.
3. **Model Training**: The AI model is trained using the data, adjusting its internal settings to improve its predictions. With more data, the model becomes more accurate and better at recognizing new examples like unseen images of cats.
4. **Decision Making**: Once trained, it can use what it has learned to make decisions. For example, it can find whether a new image contains a cat based on the patterns it learned during training.
5. **Feedback and Improvement**: It can improve through feedback, especially in methods like reinforcement learning. In this case, the AI receives rewards or penalties, refining its ability to make better decisions over time.

Types of Artificial Intelligence

AI can be classified into two main categories based on its capabilities and functionalities.

1. Based on Capabilities

- **Narrow AI**: Designed to perform specific tasks such as speech recognition or recommendation systems.
- **General AI**: A theoretical AI capable of performing human-like intellectual tasks across multiple domains.
- **Superintelligent AI**: A hypothetical AI that would surpass human intelligence in reasoning and decision-making.

2. Based on Functionalities

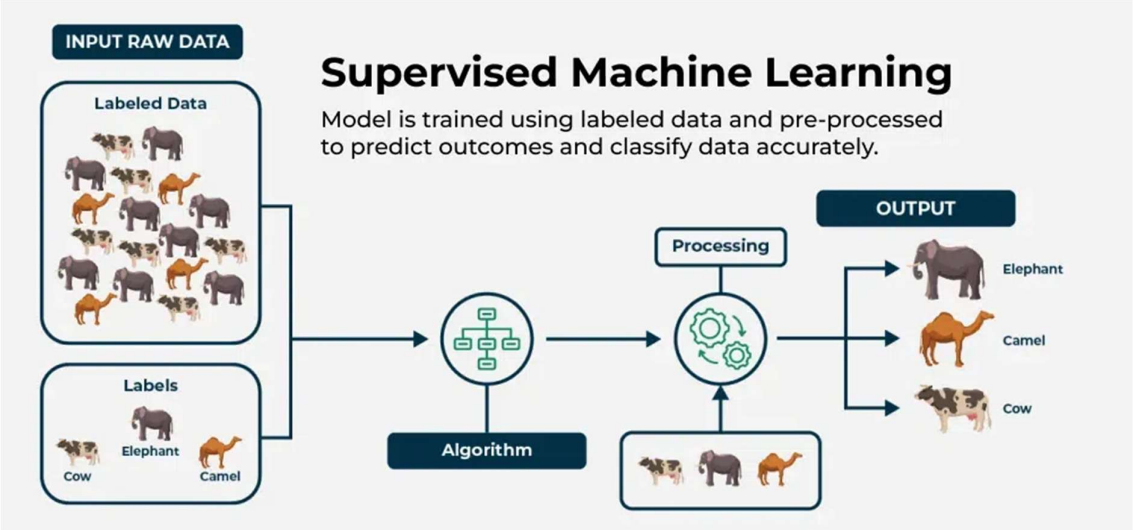
- **Reactive Machines**: AI systems that respond only to current inputs without storing past experiences.
- **Limited Memory**: AI systems that use past data and observations to improve future decisions.
- **Theory of Mind**: A theoretical AI that could understand human emotions, beliefs, and intentions.
- **Self-Aware AI**: A hypothetical AI with consciousness and awareness of its own existence.

AI Models- based on learning

AI models are systems that learn from data or predefined rules to make predictions and decisions. Different models use different learning approaches depending on the type of training data and feedback available. Based on Learning.

1. Supervised Learning Models

Supervised learning is a type of machine learning where a model learns from labelled data, meaning each input has a correct output. The model compares its predictions with actual results and improves over time to increase accuracy

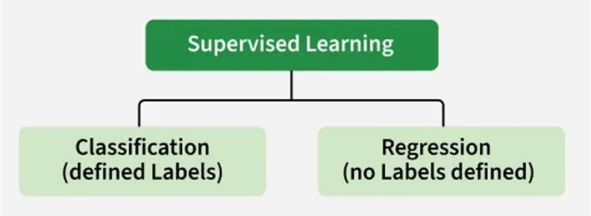


Its main features are:

- **Each input has a known output**
- **Adjusts itself to reduce prediction errors**
- **Make accurate predictions on new data**
- **For example it recognizing handwritten digits from trained data**

Types of Supervised Learning

Now, Supervised learning can be applied to two main types of problems:



- **Classification: Where the output is a categorical variable (e.g., spam vs. non-spam emails, yes vs. no).**
- **Regression: Where the output is a continuous variable (e.g., predicting house prices, stock prices).**

Let's first understand the classification and regression data through the table below:

User ID	Gender	Age	Salary	Purchased
15624510	Male	19	19000	0
15810944	Male	35	20000	1
15668575	Female	26	43000	0
15603246	Female	27	57000	0
15804002	Male	19	76000	1
15728773	Male	27	58000	1
15598044	Female	27	84000	0
15694829	Female	32	150000	1
15600575	Male	25	33000	1
15727311	Female	35	65000	0
15570769	Female	26	80000	1
15606274	Female	26	52000	0
15746139	Male	20	86000	1
15704987	Male	32	18000	0
15628972	Male	18	82000	0
15697686	Male	29	80000	0
15733883	Male	47	25000	1

Figure A: CLASSIFICATION

Temperature	Pressure	Relative Humidity	Wind Direction	Wind Speed
10.69261758	986.882019	54.19337313	195.7150879	3.278597116
13.59184184	987.8729248	48.0648859	189.2951202	2.909167767
17.70494885	988.1119385	39.11965597	192.9273834	2.973036289
20.95430404	987.8500366	30.66273218	202.0752869	2.965289593
22.9278274	987.2833862	26.06723423	210.6589203	2.798230886
24.04233986	986.2907104	23.46918024	221.1188507	2.627005816
24.41475295	985.2338867	22.25082295	233.7911987	2.448749781
23.93361956	984.8914795	22.35178837	244.3504333	2.454271793
22.68800023	984.8461304	23.7538641	253.0864716	2.418341875
20.56425726	984.8380737	27.07867944	264.5071106	2.318677425
17.76400389	985.4262085	33.54900114	280.7827454	2.343950987
11.25680746	988.9386597	53.74139903	68.15406036	1.650191426
14.37810685	989.6819458	40.70884681	72.62069702	1.553469896
18.45114201	990.2960205	30.85038484	71.70604706	1.005017161
22.54895853	989.9562988	22.81738811	44.66042709	0.264133632
24.23155922	988.796875	19.74790765	318.3214111	0.329656571

Figure B: REGRESSION

Both the above figures have labelled data set as follows:

Figure A: It is a dataset of a shopping store that is useful in predicting whether a customer will purchase a particular product under consideration or not based on his/her gender, age and salary.

- **Input: Gender, Age, Salary**
- **Output: Purchased i.e. 0 or 1; 1 means yes the customer will purchase and 0 means that the customer won't purchase it.**

Figure B: It is a Meteorological dataset that serves the purpose of predicting wind speed based on different parameters.

- **Input: Dew Point, Temperature, Pressure, Relative Humidity, Wind Direction**
- **Output: Wind Speed**

Working of Supervised Machine Learning

The working of supervised machine learning follows these key steps:

1. Collect Labeled Data

- **Gather a dataset where each input has a known correct output (label).**
- **Example: Images of handwritten digits with their actual numbers as labels.**

2. Split the Dataset

- **Divide the data into training data (about 80%) and testing data (about 20%).**
- **The model will learn from the training data and be evaluated on the testing data.**

3. Train the Model

- **Feed the training data (inputs and their labels) to a suitable supervised learning algorithm (like Decision Trees, SVM or Linear Regression).**
- **The model tries to find patterns that map inputs to correct outputs.**

4. Validate and Test the Model

- **Evaluate the model using testing data it has never seen before.**
- **The model predicts outputs and these predictions are compared with the actual labels to calculate accuracy or error.**

5. Deploy and Predict on New Data

- **Once the model performs well, it can be used to predict outputs for completely new, unseen data.**

Supervised Machine Learning Algorithms

Supervised learning includes different types of algorithms used to predict outputs based on labeled data. Each algorithm is designed for specific tasks like prediction or classification.

- **Linear Regression: Used to predict continuous values (e.g., price, temperature). It is simple and widely used.**
- **Logistic Regression: Logistic regression is a type of supervised learning classification algorithm that is used to predict a binary output variable.**
- **Decision Trees: Uses a tree-like structure where each node represents a decision and each leaf represents an outcome.**
- **Random Forests: Combines multiple decision trees to improve accuracy and reduce overfitting.**
- **Support Vector Machine(SVM): Separates data into classes using a boundary (hyperplane). Support vectors help define this boundary.**
- **K-Nearest Neighbors: Predicts based on the closest data points. Results depend on the value of k and distance measure.**
- **Gradient Boosting: Builds models step-by-step by correcting errors of previous models, creating a strong model.**
- **Naive Bayes Algorithm: Based on probability and Bayes' Theorem, assuming features are independent of each other.**

Examples

- **Fraud Detection in Banking: Uses labeled transaction data to identify and predict fraudulent activities.**
- **Parkinson Disease Prediction: Analyzes medical data such as voice recordings, motor symptoms, and clinical measurements to identify patterns and predict whether a patient has disease or not.**

- **Customer Churn Prediction: Uses historical customer data to predict whether a customer will leave a service.**
- **Cancer cell classification: Implements supervised learning for cancer cells based on their features and identifying them if they are 'malignant' or 'benign.**
- **Stock Price Prediction: Uses past data to predict stock trends and support investment decisions.**

Advantages

- **Easy to understand and implement as it learns from labeled data.**
- **Provides high accuracy when sufficient labeled data is available.**
- **Works for both classification (spam detection, disease prediction) and regression (price forecasting).**
- **Can generalize well to unseen data with proper training and diverse datasets.**
- **Widely used in applications like speech recognition, medical diagnosis, sentiment analysis and fraud detection.**

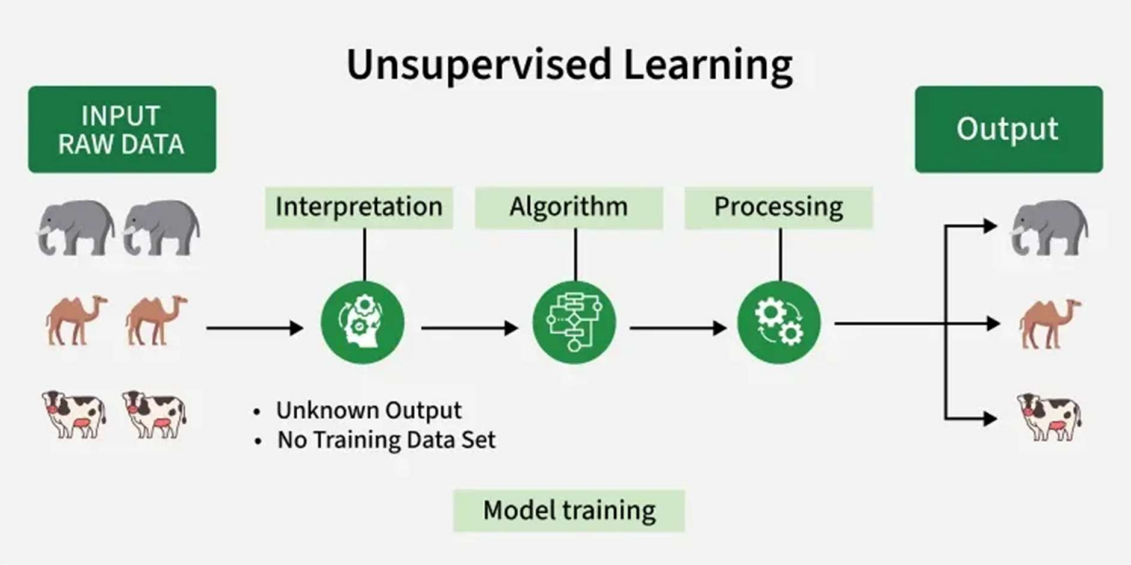
Disadvantages

- **Requires large amounts of labeled data, which is expensive and time-consuming to prepare.**
- **Can be biased if the training data is unbalanced, leading to unfair or inaccurate predictions.**
- **May overfit the training data instead of learning general patterns, especially with small datasets.**
- **Performance can drop when applied to data that is very different from the training data.**
- **Not easily scalable for problems with a very large number of labels, such as in natural language tasks.**

2. Unsupervised Learning Models

Unsupervised Learning is a type of machine learning where the model works without labelled data. It learns patterns on its own by grouping similar data points or finding hidden structures without any human intervention.

- **It is used for tasks like clustering, dimensionality reduction and Association Rule Learning.**
- **Helps identify hidden patterns in data**
- **Useful for grouping, compression and anomaly detection**



The image shows set of animals like elephants, camels and cows that represents raw data that the unsupervised learning algorithm will process.

- **The "Interpretation" stage signifies that the algorithm doesn't have predefined labels or categories for the data. It groups the data based on inherent patterns.**
- **An algorithm represents unsupervised learning process that helps to identify patterns in the data.**
- **The processing stage shows the algorithm working on the data.**

The output shows the results of the unsupervised learning process. In this case, the algorithm might have grouped the animals into clusters based on their species (elephants, camels, cows).

Working of Unsupervised Learning

The working of unsupervised machine learning can be explained in these steps:

1. Collect Unlabeled Data

- **Gather a dataset without predefined labels or categories.**
- **Example: Images of various animals without any tags.**

2. Select an Algorithm

- **Choose a suitable unsupervised algorithm such as clustering like K-Means, association rule learning like Apriori or dimensionality reduction like PCA based on the goal.**

3. Train the Model on Raw Data

- **Feed the entire unlabeled dataset to the algorithm.**

- **The algorithm looks for similarities, relationships or hidden structures within the data.**

4. Group or Transform Data

- **The algorithm organizes data into groups (clusters), rules or lower-dimensional forms without human input.**
- **Example: It may group similar animals together or extract key patterns from large datasets.**

5. Interpret and Use Results

- **Analyze the discovered groups, rules or features to gain insights or use them for further tasks like visualization, anomaly detection or as input for other models.**

Unsupervised Learning Algorithms

There are mainly 3 types of Unsupervised Algorithms that are used:

1. Clustering Algorithms

Clustering is an unsupervised machine learning technique that groups unlabeled data into clusters based on similarity. Its goal is to discover patterns or relationships within the data without any prior knowledge of categories or labels.

- **Groups data points that share similar features or characteristics.**
- **Helps find natural groupings in raw, unclassified data.**
- **Commonly used for customer segmentation, anomaly detection and data organization.**
- **Works purely from the input data without any output labels.**
- **Enables understanding of data structure for further analysis or decision-making.**

2. Association Rule Learning

Association rule learning is a rule-based unsupervised learning technique used to discover interesting relationships between variables in large datasets. It identifies patterns in the form of "if-then" rules, showing how the presence of some items in the data implies the presence of others.

- **Finds frequent item combinations and the rules connecting them.**
- **Commonly used in market basket analysis to understand product purchase relationships.**
- **Helps retailers design promotions and cross-selling strategies.**

Some common Association Rule Learning algorithms:

- ***Apriori Algorithm: Finds patterns by exploring frequent item combinations step-by-step.***
- ***FP-Growth Algorithm: An Efficient Alternative to Apriori. It quickly identifies frequent patterns without generating candidate sets.***
- ***Eclat Algorithm: Uses intersections of itemsets to efficiently find frequent patterns.***

- ***Efficient Tree-based Algorithms**: Scales to handle large datasets by organizing data in tree structures.*

3. Dimensionality Reduction

Dimensionality reduction is the process of decreasing the number of features or variables in a dataset while retaining as much of the original information as possible. This technique helps simplify complex data making it easier to analyze and visualize. It also improves the efficiency and performance of machine learning algorithms by reducing noise and computational cost.

- **It reduces the dataset's feature space from many dimensions to fewer, more meaningful ones.**
- **Helps focus on the most important traits or patterns in the data.**
- **Commonly used to improve model speed and reduce overfitting.**

Here are some popular Dimensionality Reduction algorithms:

- ***Principal Component Analysis (PCA)**: Reduces dimensions by transforming data into uncorrelated principal components.*
- ***Non-negative Matrix Factorization (NMF)**: Breaks data into non-negative parts to simplify representation.*
- ***Locally Linear Embedding (LLE)**: Reduces dimensions while preserving the relationships between nearby points.*
- ***Isomap**: Captures global data structure by preserving distances along a manifold.*

Applications

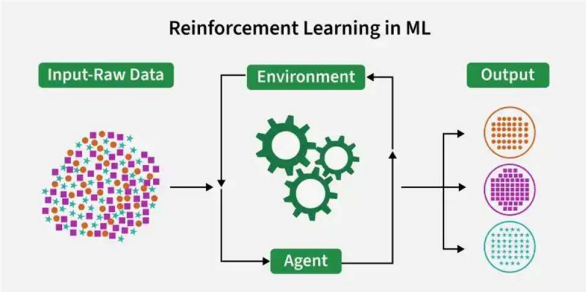
- **Customer Segmentation: Algorithms cluster customers based on purchasing behavior or demographics, enabling targeted marketing strategies.**
- **Anomaly Detection: Identifies unusual patterns in data, aiding fraud detection, cybersecurity and equipment failure prevention.**
- **Recommendation Systems: Suggests products, movies or music by analyzing user behavior and preferences.**
- **Image and Text Clustering: Groups similar images or documents for tasks like organization and content recommendation.**
- **Social Network Analysis: Detects communities or trends in user interactions on social media platforms.**

Advantages

- **Works with raw, unlabeled data, saving time and effort required for data annotation.**
- **Finds hidden patterns and natural groupings in data that may not be easily identified by humans.**
- **Handles large and complex datasets efficiently, including high-dimensional data.**
- **Helps detect anomalies and unusual data points without needing prior examples.**

3. Reinforcement Learning Mode.

Reinforcement Learning (RL) is a branch of machine learning that focuses on how agents can learn to make decisions through trial and error to maximize cumulative rewards. RL allows machines to learn by interacting with an environment and receiving feedback based on their actions. This feedback comes in the form of rewards or penalties.



Reinforcement Learning revolves around the idea that an agent (the learner or decision-maker) interacts with an environment to achieve a goal. The agent performs actions and receives feedback to optimize its decision-making over time.

- **Agent**: The decision-maker that performs actions.
- **Environment**: The world or system in which the agent operates.
- **State**: The situation or condition the agent is currently in.
- **Action**: The possible moves or decisions the agent can make.
- **Reward**: The feedback or result from the environment based on the agent's action.

Types of Reinforcements

1. Positive Reinforcement: Positive Reinforcement is defined as when an event, occurs due to a particular behavior, increases the strength and the frequency of the behavior. In other words, it has a positive effect on behavior.

- **Advantages**: Maximizes performance, helps sustain change over time.
- **Disadvantages**: Overuse can lead to excess states that may reduce effectiveness.

2. Negative Reinforcement: Negative Reinforcement is defined as (strengthening) rise of behavior because a negative condition is stopped or avoided.

- **Advantages**: Increases behavior frequency, ensures a minimum performance standard.
- **Disadvantages**: It may only encourage just enough action to avoid penalties.

Application

- **Robotics**: RL is used to automate tasks in structured environments such as manufacturing, where robots learn to optimize movements and improve efficiency.
- **Games**: Advanced RL algorithms have been used to develop strategies for complex games like chess, Go and video games, outperforming human players in many instances.
- **Industrial Control**: RL helps in real-time adjustments and optimization of industrial operations, such as refining processes in the oil and gas industry.
- **Personalized Training Systems**: RL enables the customization of instructional content based on an individual's learning patterns, improving engagement and effectiveness.

History of AI

Artificial Intelligence did not develop in one day. It developed gradually over many years through the work of mathematicians, scientists, and computer researchers.

1940s – Beginning of the Idea

Scientists started thinking about whether a **machine could behave like a human brain**.

In **1943**, Warren McCulloch and Walter Pitts developed a mathematical model of an **artificial neuron**. This became an important foundation for neural networks.

example:
A human brain has neurons that process information. Researchers wanted computers to work in a somewhat similar way.

1950 – Alan Turing

In **1950**, British mathematician **Alan Turing** asked an important question:

"Can machines think?"

He proposed the **Turing Test**.

The basic idea is that if a person communicates with a computer and cannot reliably determine whether they are communicating with a human or a machine, the machine may be considered intelligent.

Example:
Imagine chatting with two people through text. One is a human and one is a computer. If you cannot tell which is the computer, the computer has demonstrated a form of machine intelligence.

1956 – Birth of AI

The year **1956** is considered a major milestone in AI history.

At the **Dartmouth Conference**, John McCarthy and other researchers formally promoted the study of **Artificial Intelligence**.

John McCarthy is widely known for **coining the term "Artificial Intelligence."**

The main goal was:

To make machines perform tasks that require human intelligence.

1960s – Early AI Programs

During the 1960s, researchers developed programs that could perform tasks such as:

a. Solving mathematical problems

b. Playing games

c. Proving simple mathematical theorems

d. Understanding limited natural language

At this stage, AI systems mainly followed **rules programmed by humans**.

1970s–1980s – Expert Systems

Researchers began developing **Expert Systems**.

An expert system stores knowledge and rules from human experts and uses them to make decisions.

Example:
A medical expert system could use rules such as:

IF a patient has fever and certain symptoms
THEN suggest possible diseases.

These systems were useful in areas such as medicine, business, and engineering.

1980s–1990s – Machine Learning

AI gradually moved from systems that only followed fixed rules toward **Machine Learning**.

Machine learning allows a computer to **learn patterns from data**.

Example:

Instead of programming:

"If an email contains these exact words, call it spam."

We can give the computer thousands of examples of **spam and non-spam emails**, and it learns patterns to identify new spam emails.

2000s – Big Data and Better Computers

AI developed rapidly because of:

- Large amounts of digital data
- Faster computers
- Better algorithms
- Improved storage
- Internet growth

More data allowed computers to learn more complex patterns.

2010s – Deep Learning

Deep Learning, a branch of machine learning based on multi-layer neural networks, became extremely successful.

AI systems became much better at:

- Image recognition
- Speech recognition
- Language translation
- Playing games
- Understanding text

For example, computers became capable of recognizing objects in photographs and understanding spoken commands.

2020s – Generative AI

In the 2020s, **Generative AI** became widely used.

Generative AI can create new content such as:

- Text
- Images
- Audio
- Video
- Computer programs

Philosophical Foundations of AI

The **philosophical foundations of AI** are the ideas and questions that people have had about **human thinking, intelligence, knowledge, and decision-making**.

Before computers were invented, philosophers were already asking questions like:

“What does it mean to think?”
“How do humans learn?”
“Can a machine think like a human?”

These questions later became important in the development of AI.

1. Can a Machine Think?

This is one of the biggest questions in AI.

Humans can think, understand, learn, and solve problems. AI researchers wanted to know whether a computer could also do these things.

Alan Turing asked this question and proposed the **Turing Test** in 1950.

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2. Logic and Reasoning

Humans use logic to reach conclusions.

For example:

- All students have an ID card.
- Rahul is a student.
- Therefore, Rahul has an ID card.

AI also uses **logic and rules** to solve problems and make decisions.

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3. How Do Humans Get Knowledge?

Humans learn in different ways. We learn by:

- Reading
- Watching
- Listening
- Experience
- Making mistakes

Philosophers studied how humans gain knowledge. This idea became important in AI.

Today, **Machine Learning** allows computers to learn from data and experience.

Example:
If we show a computer thousands of pictures of cats and dogs, it can learn the difference between them.

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4. Human Mind and Computer

Another important question is:

"Is the human brain similar to a computer?"

The human brain receives information, processes it, stores it, and produces an answer.

Similarly, a computer:

Input → Processing → Output

This comparison helped researchers develop ideas about artificial intelligence.

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5. Turing Test

The Turing Test is a widely recognized benchmark for evaluating a machine's ability to demonstrate human-like intelligence. The core idea is simple: A human judge engages in a text-based conversation with both a human and a machine. The judge's task is to determine which participant is human and which is the machine. If the judge is unable to distinguish between the human and the machine based solely on the conversation, the machine is said to have passed the Turing Test.

The **Turing Test** is a method proposed by **Alan Turing in 1950** to check whether a machine can behave intelligently like a human.

Criteria for the Turing Test

The Turing Test does not require the machine to be correct or logical in its responses but rather to be convincing in simulating human conversation. The test is fundamentally about deception—the machine must fool the judge into believing that it is human.

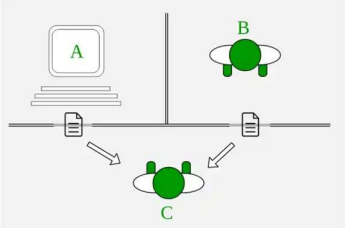
The key criteria include the following for turing test:

1. **Natural Language Processing (NLP):** The machine must understand and generate human language fluently.
2. **Knowledge Representation:** The machine needs to handle and manipulate knowledge to provide contextually relevant responses.
3. **Reasoning:** The machine should demonstrate some form of logical reasoning, even if flawed, to sustain a conversation.
4. **Learning:** Ideally, the machine should learn from the interaction, adapting its responses over time.

How the Turing Test Works?

In a typical Turing Test scenario, three participants are involved: two humans and one machine.

The interrogator, a human judge, is isolated from the other two participants. The judge asks questions to both the human and the machine, aiming to identify which one is the human. The machine's goal is to respond in a way that makes it indistinguishable from the human participant. If the judge cannot reliably identify the machine, the machine is considered to have passed the Turing Test.



Here's an example of a conversation between the interrogator and the machine:

- **Judge:** Are you a computer?
- **Machine:** No.
- **Judge:** Multiply 158745887 by 56755647.
- **Machine:** (After a long pause) [Provides an incorrect answer].
- **Judge:** Add 5,478,012 and 4,563,145.
- **Machine:** (Pauses for 20 seconds and then responds) 10,041,157.

If the judge cannot distinguish between the responses of the human and the machine, the machine passes the test. The conversation is limited to a text-only format, such as a computer keyboard and screen, to prevent the judge from being influenced by any non-verbal cues.

Advantages of the Turing Test

1. Evaluates Machine Intelligence

The Turing Test provides a simple way to check whether a machine can **behave and communicate like a human**.

Example:
If a chatbot can have a natural conversation with a person, it may show human-like conversational ability.

2. Provides a Benchmark

It gives AI researchers a **standard or goal** for developing intelligent machines.

Example:
Researchers can try to develop machines that can communicate so naturally that humans cannot easily distinguish them from people.

3. Encourages AI Research

The Turing Test has encouraged researchers to work on areas such as:

- Natural Language Processing
- Reasoning
- Knowledge representation
- Machine learning

This has contributed to the development of more advanced AI systems.

4. Easy to Conduct

The test is relatively simple to perform.

It mainly requires:

- A human judge
- A human participant
- A machine
- Text-based communication

Disadvantages of the Turing Test

1. Limited Scope

The Turing Test mainly checks **conversation and language**.

It does not properly test other abilities such as:

- Vision
- Physical movement
- Problem-solving
- Decision-making
- Emotional understanding

Example:
A machine might be excellent at chatting but unable to perform a physical task.

2. Human Bias

The result depends on the **human judge**.

Different judges may have different opinions about whether a machine sounds human.

Example:
One judge may think a chatbot is human, while another judge may easily identify it as a machine.

3. Does Not Prove Real Intelligence

Passing the Turing Test does not necessarily mean that the machine **actually understands or thinks like a human**.

A machine may simply be very good at **imitating human conversation**.

This is one of the main ideas behind **John Searle's Chinese Room Argument**.

4. Not Suitable for All Real-World AI

Real-world AI systems often need abilities beyond conversation.

For example:

- Self-driving vehicles need perception and decision-making.
- Medical AI needs analysis and diagnosis support.
- Robots need to understand and interact with the physical world.

Chinese Room Argument

Searle's Chinese Room is a thought experiment proposed by John Searle in 1980 to argue that a computer can process symbols and give intelligent-looking answers without actually understanding them.

Simple Example

Imagine:

- **A person is sitting inside a room.**
- **The person does not know Chinese.**
- **People outside the room send questions written in Chinese.**
- **Inside the room, the person has a rule book that tells them which Chinese symbols to return.**
- **By following the rules, the person sends correct Chinese answers.**
- **People outside think: "The person inside understands Chinese."**
- **But actually, the person does not understand Chinese—they are only manipulating symbols.**

Main Idea

Following rules is not the same as understanding.

Searle used this example to challenge the idea of Strong AI, which claims that a computer running the right program can actually have a mind and understanding.

Chinese Room and AI

Computer/AI	Human in Room
Receives input	Receives Chinese symbols
Processes symbols	Follows rules
Produces output	Produces Chinese answer
May appear intelligent	Appears to understand Chinese
Does not necessarily understand	Actually does not understand

Searle's Argument

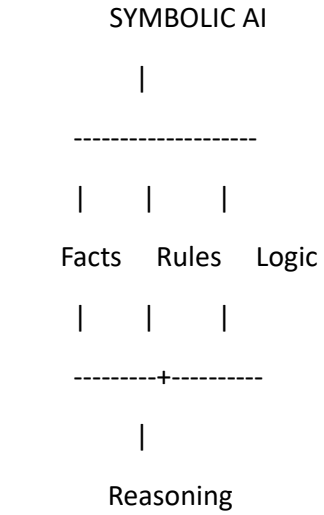
1. **Computers manipulate symbols according to rules.**
2. **Manipulating symbols does not automatically create understanding.**
3. **Therefore, a computer can produce intelligent answers without truly understanding what it is doing.**

Strong AI vs Weak AI

- **Strong AI: A properly programmed computer can genuinely understand and have a mind.**
- **Weak AI: Computers can be useful tools for simulating or performing intelligent behavior.**

Symbolic Ai

Symbolic AI is an approach to Artificial Intelligence in which knowledge is represented using **symbols, facts, rules, and logic**, and the system uses **reasoning** to solve problems and make decisions.



|

Conclusion

Example-

If a person have -

Fever + Cough
↓
Apply Rule
↓
Possible Flu

Suppose that I give to things to computer

Fact: Ravi is a student.

Rule: If a person is a student, then the person studies.

Therefore:

Ravi is a student → Ravi studies.

Advantages

1. **Easy to understand** – Rules are clear and understandable.
2. **Explainable** – The system can explain how it reached a conclusion.
3. **Good for logical reasoning** – Useful for solving logic-based problems.
4. **No large dataset required** – Knowledge can be directly given as rules.
5. **Easy to modify** – New rules can be added or existing rules can be changed.
6. **Useful in expert systems** – Used in areas such as diagnosis, planning, and decision-making.

Disadvantages

1. **Rule creation is difficult** – Humans have to create many rules manually.
2. **Poor handling of uncertainty** – Real-world situations are often uncertain. (it may rain tomorrow).
3. **Difficult to handle complex data** – Images, speech, and unstructured data are difficult to represent.
4. **Too many rules become complex** – Large systems can have thousands of rules.
5. **Limited learning ability** – Traditional Symbolic AI does not automatically learn patterns like Machine Learning.
6. **Maintenance can be difficult** – Updating a large knowledge base can require significant effort.

Connectionist Ai

Connectionist AI is an approach to Artificial Intelligence that uses interconnected artificial neurons to learn patterns from data and make decisions or predictions.

Many Examples

↓

Neural Network

↓

Learns Patterns

↓

New Input

↓

Prediction

Another Simple Example: Face Recognition

Suppose we want an AI system to recognize **Rahul's face**.

We provide many images:

Rahul's photo 1

Rahul's photo 2

Rahul's photo 3

Rahul's photo 4

↓

Neural Network

↓

Learns facial patterns

The network learns features such as

Eyes, Nose ,Face shape ,Distance between eyes ,Other visual patterns

Example: Email Spam Detection

Suppose we want AI to identify whether an email is **Spam** or **Not Spam**.

We give the system thousands of emails.

Training data

Email	Result
"Win ₹10,000 now!"	Spam
"Congratulations! You won!"	Spam
"Meeting at 10 AM"	Not Spam
"Please send the report"	Not Spam

The neural network learns patterns from these examples.

For example, it may learn that words such as:

- "Win"
- "Prize"
- "Free"
- "Congratulations"

are often associated with spam.

Then a new email arrives:

"Congratulations! You have won a free prize."

The network processes it and may produce:

Spam probability = 98%

Not Spam probability = 2%

Therefore:

Output → SPAM

Advantages of Connectionist AI

1. Learns from examples

It can learn patterns from training data instead of requiring every rule to be manually programmed.

2. Good for pattern recognition

It is very useful for:

- Face recognition
- Speech recognition
- Image recognition
- Handwriting recognition

3. Handles complex problems

Neural networks can learn complicated relationships between inputs and outputs.

4. Improves with suitable training data

With appropriate data and training, the model can improve its predictions.

5. Useful in many applications

Used in:

- Chatbots
- Medical diagnosis
- Recommendation systems
- Self-driving technologies
- Voice assistants

Cons of Connectionist AI:

1. **Interpretability:** Connectionist AI systems are often seen as "black boxes" due to their lack of transparency and interpretability.
2. **Resource Intensive:** Training neural networks requires significant computational resources and large datasets.
3. **Overfitting:** These systems are prone to overfitting, where they perform well on training data but poorly on new, unseen data.

Key Differences between Symbolic and Connectionist AI

Feature	Symbolic AI	Connectionist AI
Knowledge Representation	Uses explicit symbols and rules	Uses connections and weights in networks
Transparency	High, interpretable and traceable	Low, often considered a "black box"
Reasoning Ability	Excels in logical reasoning and problem-solving	Excels in pattern recognition and prediction
Scalability	Struggles with large, complex datasets	Scales well with large datasets

Feature	Symbolic AI	Connectionist AI
Flexibility	Less flexible and adaptable	Highly adaptable through training
Learning Method	Relies on hand-coded rules	Learns from data through training
Resource Requirements	Generally less resource-intensive	Requires significant computational resources
Risk of Overfitting	Lower risk	Higher risk

Which One to Use?

Choosing between Symbolic AI and Connectionist AI depends on the specific application and requirements:

- **Symbolic AI:** Suitable for tasks requiring explicit knowledge representation, logical reasoning, and interpretability. Examples include expert systems, theorem proving, and rule-based systems.
- **Connectionist AI:** Ideal for tasks involving large datasets, pattern recognition, and adaptability. Examples include image and speech recognition, natural language processing, and predictive analytics.

Concept of Intelligent Agents.

In the realm of AI, Intelligent Agents drive automation and decision-making with cognitive abilities.

- Intelligent agents represent a subset of AI systems demonstrating intelligent behaviour, including adaptive learning, planning, and problem-solving.
- It operate in dynamic environments, where it makes decisions based on the information available to them.
- These agents dynamically adjust their behaviour, learning from past experiences to improve their approach and aiming for accurate solutions.

The design of an intelligent agent typically involves four key components:

1. **Perception:** Agents have sensors or mechanisms to observe and perceive aspects of their environment. This may involve collecting data from the physical world, accessing databases, or receiving input from other software components.

2. **Reasoning:** Agents possess computational or cognitive capabilities to process the information they perceive. They use algorithms, logic, or machine learning techniques to analyze data, make inferences, and derive insights from the available information.
3. **Decision-Making:** Based on their perception and reasoning, agents make decisions about the actions they should take to achieve their goals. These decisions are guided by predefined objectives, which may include optimizing certain criteria or satisfying specific constraints.
4. **Action:** Agents execute actions in their environment to affect change and progress towards their goals. These actions can range from simple operations, such as sending a message or adjusting parameters, to more complex tasks, such as navigating a virtual world or controlling physical devices.

Examples of Intelligent Agents include self-driving cars, recommendation systems, virtual assistants, and game-playing AI.

Rational Agents and Rationality in Decision-Making

Intelligent agents are characterized by their rationality in decision-making, which aims to attain optimal outcomes or, in uncertain scenarios, the best-expected outcome.

A rational agent can be said to those, who do the right thing, It is an autonomous entity designed to perceive its environment, process information, and act in a way that maximizes the achievement of its predefined goals or objectives. Rational agents always aim to produce an optimal solution.

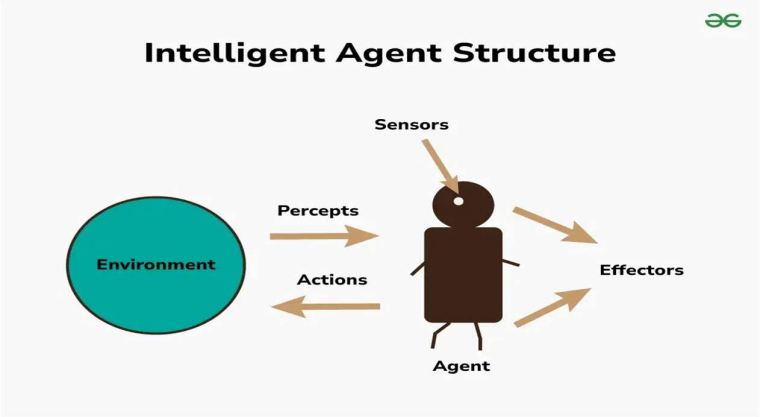
Rationality in AI refers to the principle that such agents should consistently choose actions that are expected to lead to the best possible outcomes, given their current knowledge and the uncertainties present in the environment. This principle of rationality guides the behavior of intelligent agents in the following ways:

- **Perception and Information Processing:** Rational agents strive to perceive and process information efficiently to gain the most accurate understanding of their environment.
- **Reasoning and Inference:** They employ logical reasoning and probabilistic inference to make informed decisions based on available evidence and prior knowledge.
- **Decision-Making Under Uncertainty:** When faced with uncertainty, rational agents weigh the probabilities of different outcomes and choose actions that maximize their expected utility or achieve the best possible outcome given the available information.
- **Adaptation and Learning:** Rational agents adapt their behavior over time based on feedback and experience, continuously refining their decision-making strategies to improve performance and achieve their goals more effectively.

Example of a rational agent is a chess-playing AI, which selects moves with the highest likelihood of winning.

How Intelligent Agent work Inside?

An agent's internal workings involve Agent program that run on computing device and process the data comes from the environment through its architecture. Let's discuss how an agent works from the inside using program and architecture:



1. **Environment:** Environment is the area around the agent that it interacts with. An environment can be anything like a physical space, a room or a virtual space like a game world or the internet.
2. **Sensors:** Sensors are tools that AI agent uses to perceive their environment. They can be any physical like cameras, microphones, temperature sensors or a software sensor that read data from files.
3. **Actuators:** Actuators are tools that AI agent uses to interact with their environment through some actions. They can be any physical actuators like wheels, motors, robotic hands, or computer screens or they can be software actuators that send messages.
4. **Effectors:** Effectors take instructions from decision making mechanism and translates them into actions and these actions are performed through actuators.

2. Program or Decision-making mechanism:

This is the brain of the AI agent, this mechanism processes the information that is received through sensors and makes decisions through that data using programs. Let's understand how the agent's program execute the operations.

- The decision-making mechanism, often referred to as the agent's program, processes information from sensors and makes decisions based on that data.
- The program takes current percepts as input and generates actions for the actuators.
- It embodies the agent function, which maps percepts to actions based on the agent's goals and objectives.
- Various **types of agent programs** exist, such as simple reflex agents, model-based reflex agents, goal-based agents, and utility-based agents.
- These programs differ in how they process percepts and generate actions, depending on the agent's design and task requirements.

For example, a simple reflex agent may have a program that directly maps percept states to actions without considering past or future percepts for a two-state vacuum environment. This decision will be executed through effectors.

PEAS Representation of AI agent

PEAS stands for performace measure, environment, actuators and sensors. It is a framework that is used to describe an AI agent. It's a structured approach to design and understand AI systems.

1. **Perfromance measure:** Performance measure is a criteria that measures the success of the agent. It is used to evaluate how well the agent is acheiving its goal.
   For example, in a spam filter system, the performance measure could be minimizing the number of spam emails reaching the inbox.

2. **Environment**: The environment represents the domain or context in which the agent operates and interacts. This can range from physical spaces like rooms to virtual environments such as game worlds or online platforms like the internet.

3. **Actuators**: Actuators are the mechanisms through which the AI agent performs actions or interacts with its environment to achieve its goals. These can include physical actuators like motors and robotic hands, as well as digital actuators like computer screens and text-to-speech converters.

4. **Sensors:** Sensors enable the AI agent to gather information from its environment, providing data that informs its decision-making process and actions. These sensors can capture various environmental parameters such as temperature, sound, movement, or visual input. Examples of sensors include cameras, microphones, temperature sensors, and motion sensors.

Applications of Intelligent Agents

Intelligent agents find applications across a wide range of domains, revolutionizing industries and enhancing human capabilities. Some notable applications include:

1. **Autonomous Systems:** Intelligent agents power autonomous vehicles, drones, and robots, enabling them to perceive their surroundings, navigate complex environments, and make decisions in real-time.

2. **Personal Assistants:** Virtual personal assistants like Siri, Alexa, and Google Assistant employ intelligent agents to understand user queries, retrieve relevant information, and perform tasks such as scheduling appointments, setting reminders, and controlling smart home devices.

3. **Recommendation Systems:** E-commerce platforms, streaming services, and social media platforms utilize intelligent agents to analyze user preferences and behavior, providing personalized recommendations for products, movies, music, and content.

4. **Financial Trading:** Intelligent agents are employed in algorithmic trading systems to analyze market data, identify trading opportunities, and execute trades autonomously, maximizing returns and minimizing risks.

Challenges for Intelligent Agents

Despite their immense potential, intelligent agents also pose several challenges and considerations:

1. **Ethical and Legal Implications:** Intelligent agents raise ethical concerns regarding privacy, bias, transparency, and accountability. Developers must ensure that agents behave ethically and comply with legal regulations and societal norms.
2. **Robustness and Reliability:** Agents must be robust and reliable in dynamic and uncertain environments. They should be capable of handling unexpected situations, adversarial attacks, and noisy or incomplete data.
3. **Interpretability:** Understanding and interpreting the decisions made by intelligent agents is crucial for building trust and transparency. Explainable AI techniques are essential for providing insights into the reasoning process and decision-making of agents.
4. **Scalability and Efficiency:** As AI systems become increasingly complex and data-intensive, scalability and efficiency become critical considerations. Designing agents that can scale to large-scale deployments and operate efficiently with limited computational resources is essential.